Filippo Cenacchi

Doctoral Researcher (Computer Science) & Product Manager
Academic profile - Linked-in profile

Email: filippo.cenacchi@hdr.mq.edu.au
Mobile: +61 450 241 740

Profile

Award-winning Ph.D candidate in Computer Science with publications at top-tier AI and computing venues, extensive university teaching experience and a proven background in industry product leadership. I lead research that (1) advances multimodal AI and reliable machine learning, (2) develops human–AI interaction and intelligent healthcare systems, (3) translates complex domain knowledge into novel computational architectures, and (4) delivers rigorous, auditable and impactful real-world solutions. I combine research leadership with industry execution: defining ambitious research directions, building interdisciplinary collaborations, rapidly developing and validating new methods, mentoring and teaching students, and translating technical advances into measurable scientific and societal impact.

Education

- **PhD, Computer Science** — Sydney, AU
  Macquarie University (Ranked Top-50 globally for Data Science and AI, QS 2025) — *2025 – 2027*
  - **Thesis**: A Computational Framework for Reliable, Interactive, and Auditable Patient Digital Twins shaped by Humanoid AI agents.
- **MSc, Situated Interactions (Excellent with Honors)** — Paris, FR
  Université Paris-Saclay (Ranked #1 in France #13 globally, ARWU 2025) — *2023 – 2025*
  - **Focus**: Situated interaction design combining human, technical, and organizational constraints.
- **MSc, Human-Computer Interactions (Excellent)** — Stockholm, SE
  KTH Royal Institute of Technology (Ranked #1 in Sweden, Top-10 in EU for CS, THE 2024) — *2022 – 2024*
  - **Focus**: Human-centered design, HCI methods, and applied AI/innovation.
- **BSc, Computer Science** — Bolzano, IT
  Free University of Bozen-Bolzano; Erasmus at University of Helsinki — *2019 – 2022*
  - **Focus**: Software engineering and computational methods; multilingual academic training.

Selected Research Contributions

Explore in detail →

- **Counterfactual Fragility Certificates: Exposing Hidden AI Failures in Cancer Detection**: Addressed a consequential gap in AI reliability: high-confidence predictions can become unsupported when evidence is missing or degraded. **Contribution:** designed and implemented a model-agnostic audit method that traces prediction fragility under grouped evidence removal and degradation. Across seven tabular benchmarks and diverse model families, achieved 0.915 AUROC for detecting brittle high-confidence cases, capturing 88.9% within a 20% review budget. **Application and impact:** applied in a large-scale pancreatic cancer detection project involving five NSW hospitals, providing an inspectable method for examining predictions that confidence scores alone may fail to flag. [Paper] [Code]
- **SEB-Cal: Improving Time-Series Reliability for Epidemic Forecasting and Analytics**: Addressed overconfident time-series predictions by assessing whether confidence is supported by the underlying signal. **Contribution:** developed a post-hoc method combining output confidence with spectral descriptors while preserving predicted labels, with validation-based selection and fallback to an output-space baseline. Evaluated across eight UCR/UEA datasets and eight backbone families; reduced FalseConf@0.9 from 0.128 to 0.094 on the matched evaluation subset. **Adoption and impact:** currently integrated into the main platform used by the Australia–Aotearoa Consortium for Epidemic Forecasting & Analytics (ACEFA), translating the method into an operational forecasting and analytics workflow. [Paper] [Code]
- **Passive Dementia Screening: From Facial Micro-Dynamics to Humanoid-Assisted Assessment**: Investigated whether facial behaviour can reveal cognitive risk without requiring speech or transcripts, supporting assessment where cognitive decline may overlap with depression or chronic pain. **Contribution:** developed an interpretable video-analysis pipeline extracting stabilised blink, gaze, mouth–jaw and head-motion time series. Achieved 0.953 AUROC and 0.851 F1 on 300 YT-DemTalk clips with self-reported dementia labels. **Clinical application and impact:** used within a 50-patient clinical study involving conversational humanoid AI to investigate possible early dementia indicators alongside depressive and chronic pain symptoms (*Autonomous Humanoid Robot for Chronic Pain and Depression Screening*, project 520251946065262). This extended the pipeline from retrospective video analysis to patient-facing clinical research; prospective diagnostic validity remains to be established. [Paper] [Project] [Code]

Selected A*/A Research

- **The Decision Twin: Metaverse Patient Digital Twins as Executable Clinical Reality**: KDD 2026.
- **Personalised AI coaching technology: a systematic mapping review**: ACM TIIS, 2026.
- **VR & Work: Implications of Facial Representation in VR**: Journal of Strategic Information Systems, 2025.

Invited Talks

- **Can humanoid robots help diagnose cognitive and affective health?**: Invited talk at School of Computing Industry Networking Event (Macquarie Masonic Centre), 21 Nov 2025.

Prizes & Grants

- **Macquarie University Research Excellence Scholarship iMQRES (ID:20246179)**: (Grant: $117k), 2025.
- **Macquarie University 3MT**: School of Computing 3MT: 2nd Place & People's Choice (Grant: $600), 2025, 2026.
- **Macquarie University 3MT**: Faculty of Science & Engineering 3MT: 2nd Runner Up (Grant: $500), 2025, 2026.
- **Macquarie University 3MT**: University 3MT: 2nd Runner Up (Grant: $350), 2025, 2026.
- **PERC Best Research Award**: Recognition at Performance and Expertise Research Centre Conference (Macquarie University), Sydney, 2025.

Leadership & Supervision

- **Graduate Research Student Representative**: Faculty Research Training Committee, Macquarie University (Jul 2025 – Present): representing doctoral and master's research students in training quality, policy, and student support initiatives.
- **Summer Internship Supervisor**: Macquarie University (Oct 2025 – Present): supervising summer interns in humanoid AI, multimodal fusion, and simulation-based training (Master students: Chieri Ishikawa, Tuong Vy Vu).

Research Portfolio

Clinical AI & Patient Digital Twins

- **Passive Dementia Screening via Facial Temporal Micro-Dynamics Analysis of In-the-Wild Talking-Head Video**: Pattern Recognition, under review, 2026; arXiv:2511.13802.
- **SimClinician: An Auditable Multimodal Interface for Psychologist–AI Collaboration in Mental Health Diagnosis**: HEALTH, Manuscript ID: HEALTH-2026-0066, under review, 2026.
- **Tri-Modal Severity-Fused Diagnosis across Depression and PTSD**: IEEE Transactions on Affective Computing (TAFFC), under review, 2026.
- **Humanoid AI Robots in Early Disease Detection: Challenges, Techniques and Opportunities**: ACM Computing Surveys (CSUR), Manuscript ID: CSUR-2025-0774, under review, 2025.

Reliable Machine Learning & Evidence Auditing

- **Spectral Evidence Bundling at Classification: Auditable Frequency-Energy Tracebacks for Per-Sample Prediction**: AAAI, under review, 2027.
- **FALCON-Cert: Shared Witness Design for Complete Review Audits**: AISTATS, under review, 2027.
- **Counterfactual Fragility Certificates for Tabular Classifications: Guarantees and Limits**: ICLR, under review, 2027.
- **Evidence Contracts for Domain-Mapped Prediction**: ICLR, under review, 2027.
- **What Does a Predictive Gain Certify? Auditing Information Claims under Approximate Inference**: ICLR, under review, 2027.
- **Training Scores for Conditional Audits: Overlap Weighting and Power under Confounding**: ICLR, under review, 2027.

Representation Learning & Multimodal Identifiability

- **Sparse-by-Design Cross-Modality Prediction: L0-Gated Representations for Reliable and Efficient Learning**: arXiv:2603.26801, 2026.
- **Which Tasks Identify a Representation? Specificity Bounds and Task Acquisition**: ICLR, under review, 2027.
- **AcousticPose-ID: What Speaker Motion Is Identifiable from Speech Alone? Identifiability, Negative Controls, and Cross-Corpus Evidence**: AAAI, under review, 2027.

Human–AI Interaction & Embodied Virtual Environments

- **Perceptions of Facial Misrepresentation in Virtual Reality**: International Journal of Human–Computer Studies, 2026.
- **To Be or Not To Be: Respecting when Socially Interactive Agents Should or Should Not Conform to Expectations**: Australasian Computer Science Week (ACSW), 2026.
- **Beyond the Screen: Leveraging Virtual Reality and Socially Interactive Agents to Redefine Bias Mitigation**: Journal on Multimodal User Interfaces (JMUI), under review, 2025.
- **RealityBind: Cross-Reality Event Mining for Embodied Generative AI**: AAMAS Blue Sky, under review, 2026.

Collective Decisions & Certified Agent Evaluation

- **When Agreement Backfires: Support Monotonicity in Learned-Group Consensus**: AAMAS, under review, 2027.
- **Safe Executions Can Hide Policy Failure: A Proposal-Level Audit of Hybrid Control in Synthetic Interviews**: ICLR, under review, 2027.
- **How Much Human Review Does a Certified Web-Agent Leaderboard Need?**: ACM The Web Conference (WWW), long-paper submission, 2027.
- **The Price of Certifying Cohort Recommendations: Budgeted Slate Identification**: ACM The Web Conference (WWW), long-paper submission, 2027.

Teaching Experience

Taught, supervised, assessed, and supported approximately 686 cumulative enrolments across eight undergraduate and postgraduate computing units at Macquarie University, spanning artificial intelligence, programming, databases, cybersecurity, data science, virtual reality, and industry-based projects.

- **Senior Teaching Assistant, COMP3300: Data Privacy and Information Security**: 2026; coordinated teaching support and guided students in privacy-by-design, risk assessment, differential privacy, threat analysis, data assurance, and secure data management.
- **Teaching Assistant and Workshop Instructor, COMP8410: Advanced Topics in Artificial Intelligence**: 2026; delivered master's-level teaching on generative models, reinforcement learning, diffusion models, non-IID learning, and humanoid modelling; approximately 70 students.
- **Project Supervisor, COMP8851: Major Project for Master of Data Science**: 2025–2026; supervised master's-level capstone projects across research design, technical development, experimental evaluation, documentation, and final delivery; approximately 21 enrolled students.
- **Senior Teaching Assistant, COMP3850: Computing Industry Project**: 2025–2026; Supported project-based learning, requirements analysis, teamwork, stakeholder communication, technical delivery, assessment, and detailed feedback across a unit enrolling approximately 300 students.
- **Teaching Assistant, COMP3310: Secure Applications Development**: 2025–2026; Taught secure-development workflows, common application vulnerabilities, threat mitigation, and robust engineering decision-making; approximately 35 students.
- **Teaching Assistant, COMP2350: Database Systems**: 2025–2026; Taught relational database design, SQL querying, transaction management, data modelling, and database implementation using MySQL; approximately 85 students.
- **Teaching Assistant, COMP1000: Introduction to Computer Programming**: 2025–2026; Supported approximately 90 students in computational thinking, program design, logic, debugging, and foundational programming practice.
- **Assessment & Tutoring, COMP3180: Virtual Reality and Advanced Game Development**: Marked and provided feedback on student research proposals, with attention to project framing, research design, clarity, and feasibility; approximately 40 students.

Intellectual Property & Invention Disclosure

- **Domain-Knowledge-Governed Multimodal Machine Learning Architecture for Auditable Severity Estimation and Evidence-Bundled Human Review**: Filippo Cenacchi; inventor. Macquarie University, MQ Ref. 2026-030, 2026.
- **Depth-Camera Surface-Normal Correction of Remote Photoplethysmography for Passive Heart-Rate Sensing, with Barometric Stair-Climbing and Keyboard-Kinematics Corroboration**: Filippo Cenacchi; inventor. Macquarie University, MQ Ref. 2026-041, 2026.
- **Support-Gated Causal Digital Twin for Individually-Calibrated "What-If" Health Simulation**: Filippo Cenacchi; inventor. Macquarie University, MQ Ref. 2026-042, 2026.
- **Identifiability-Gated Attribution of Digital Twin Body-Shape Drift with Minimal-Query Disambiguation**: Filippo Cenacchi; inventor. Macquarie University, MQ Ref. 2026-043, 2026.
- **Deriving Region-Specific Physiological Age Estimates from Camera-Based Motion Capture of a Structured Multi-Step Exercise Protocol, Using Clinically-Validated Age-Normative Reference Curves with Slope-Propagated Measurement Uncertainty**: Filippo Cenacchi; inventor. Macquarie University, MQ Ref. 2026-044, 2026.
- **Riemannian Hyperbolic Embedding of a Hierarchical Body-System Graph with Precision-Driven Adaptive Dimensionality and Geodesic Exercise Recommendation via Parallel-Transported Multi-Region Improvement Vectors**: Filippo Cenacchi; inventor. Macquarie University, MQ Ref. 2026-045, 2026.

Academic Service

- **Session Chair: ACM SIGKDD 2026**: AI for Sciences Track; Jeju, Republic of Korea, 9–13 August 2026.
- **Conference Reviewer: KDD 2026, AAAI 2026-2027, WWW 2027, ICLR 2027, AISTATS 2027, AAMAS 2027, IJCAI 2025-2026**: Special Track on AI4Tech: AI Enabling Critical Technologies, 2025, 2026.
- **Conference Volunteer: IEEE SERVICES 2026**: 2026 IEEE World Congress on Services; Macquarie University, Sydney, Australia, 13–18 July 2026.
- **Conference Volunteer: Australian Data Science Week**: Supported event delivery and participant engagement, 2026.
- **Session Chair & Volunteer: PAKDD 2025**: 29th Pacific-Asia Conference on Knowledge Discovery and Data Mining; Sydney, Australia, 10–13 June 2025.

Skills Summary

- **Machine Learning & Deep Learning**: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM; supervised and unsupervised learning; neural networks; model training, tuning and evaluation.
- **Generative AI & NLP**: Hugging Face Transformers, LangChain, OpenAI APIs; large language models, embeddings, retrieval-augmented generation (RAG), prompt engineering and semantic search.
- **Data Engineering & Analytics**: Python, SQL, Pandas, NumPy, SciPy, Apache Spark; data ingestion and cleaning; feature engineering; entity resolution; automated extraction from semi-structured sources; reproducible pipelines.
- **Computer Vision & Multimodal AI**: OpenCV, TorchVision, CNNs, vision transformers; image, speech and multimodal data processing; feature-level and decision-level fusion.
- **MLOps & Deployment**: MLflow, Docker, FastAPI, Git, GitHub Actions; experiment tracking, model versioning, REST API development, containerisation, CI/CD and production monitoring.

- **Cloud & Databases**: AWS, Azure, PostgreSQL, MySQL, MongoDB; cloud-based model training, data storage and scalable deployment.
- **Evaluation & Responsible AI**: Outcome frameworks; KPI definition; baseline-versus-uplift reporting; cross-validation; ablation studies; explainability; bias analysis; validation and audit trails.
- **Dashboards & Reporting**: Power BI, Tableau, Excel, Matplotlib, Seaborn, Plotly; stakeholder-facing dashboards, executive-ready reporting and metric storytelling.
- **Product Delivery**: Agile and Scrum; requirements workshops, backlog and sprint execution, stakeholder alignment, technical documentation and handover.
- **Research Strengths**: Experimental design; statistical analysis; mixed-methods evaluation; literature synthesis; rapid prototyping and fast iteration under ambiguity.

Experience (5+ years)

- **Macquarie University (industry collaboration with CTC Queensland)** Sydney, AU
  Research Assistant *Mar 2024 – Jan 2025*
  - **Requirements & Metrics**: Ran stakeholder-driven requirements capture and translated program goals into measurable outcomes and reporting needs.
  - **Data-to-Insights Delivery**: Built analysis workflows that converted participant and operational data into actionable insights for prototype refinement and reporting.
  - **Validation**: Designed evaluation runs and synthesized results into clear recommendations for decision-makers.
- **Meta** Sydney, AU
  Researcher (Contract) *Feb 2024 – Jan 2025*
  - **Ethical VR**: Ethical VR research in Macquarie partnership; user-to-user communication studies; recommendations for safer and more inclusive virtual interaction design.
- **Genvalues.com** Stockholm, SE (Hybrid)
  Product Manager *Oct 2022 – Jan 2024*
  - **Insight Delivery**: Built KPI reporting loops from user/product data to guide roadmap decisions and communicate impact to internal and B2B stakeholders.
  - **Execution**: Owned end-to-end delivery: requirements, prioritization, sprint planning, stakeholder alignment, and release documentation.
- **IamIP** Stockholm, SE
  Associate Product Manager *Mar 2022 – Sep 2022*
  - **MVP Delivery**: Coordinated MVP feature delivery, client pilots, and sprint execution with enterprise stakeholders.
- **Würth Phoenix** Bolzano, IT
  Project Manager *Jun 2021 – Mar 2022*
  - **Enterprise Delivery**: Led stakeholder-facing delivery of data-driven platform solutions (requirements, testing, rollout, documentation) in regulated, customer-critical settings.
- **Smart VR Lab** Amsterdam, NL
  VR Developer / Project Manager *Mar 2020 – Jun 2021*
  - **VR Products**: Developed VR training products and platform APIs; supported CRM rollout and medical training experiences.

Licenses & Certifications

- **Strategising: Management for Global Competitive Advantage (Specialization) (Macquarie University)**: Issued Jul 2024 Credential ID: 3SBZUD34JHWU
- **Developing an Agile Team (University of Colorado)**: Issued Jun 2024 Credential ID: ARHLT65CD8R2
- **Budgeting and Scheduling Projects (University of California, Irvine (Paul Merage School of Business))**: Issued Jun 2024 Credential ID: JVVPJKEX5W7P
- **Storytelling and Influencing: Communicate with Impact (Macquarie University)**: Issued May 2024 Credential ID: 6B8QRC3ZKTMT
- **Advanced Competitive Strategy (Ludwig-Maximilians-Universität München)**: Issued May 2024 Credential ID: ZMCJ78UUU9BM
- **Excel Fundamentals for Data Analysis (Macquarie University)**: Issued Apr 2024 Credential ID: 353FUA3QNZLU
- **Nova Membership Certificate (Top 3% Talent Among 2022 graduates Italy) (Nova)**: Issued Oct 2022 Credential ID: 2W8KBSXMO66BV78ZNSI4ZG3KJ
- **IELTS Academic Certificate C1+ (IELTS Official)**: Credential ID: A3-IT010-S-12590198